PRACTICAL 12

Event Management CRUD with Poster Upload and File Validation

AIM

To develop an admin event management module in StudentHub with add, view, edit and delete operations, along with secure poster upload and file validation.

Introduction

The admin dashboard provides a Manage events page. Event details and poster filenames are stored in MySQL. The form validates the event fields and uploaded image before saving. Category and open/closed filters, random image names, upload directory protection and an audit log are also included.

Tools and Technology

VS Code, XAMPP, PHP, MySQLi, MariaDB, Bootstrap 5.3.3, HTML, CSS, JavaScript, browser, Git and GitHub.

Files Used

File	Purpose
events-manage.php	Event form, list, edit and confirmed deletion.
php/event-functions.php	Prepared queries, validation, posters and audit log.
event-poster.php	Displays a poster using its database event ID.
php/get-events.php	Loads database events for the campus event page.
css/event-management.css	Event form and list styling with Bootstrap.
sql/practical12.sql	Event columns, sample records and audit table.
sql/practical12-events-dump.sql	Events table structure and sample data.
scripts/setup-practical12.php	Creates the protected poster directory.
scripts/test-practical12.py	Repeatable CRUD and upload validation tests.

Key Skills Used

PHP-MySQL CRUD, prepared statements, image upload, server-side validation, event status, admin access, filters and feedback messages.

Implementation Overview

1 Database and Admin Access

The events table stores title, description, event date, venue, category, status and optional poster filename. The module is linked from the admin dashboard. Anonymous visitors are redirected to login and student accounts are denied access.

2 Add, View and Edit Events

The form checks title, venue, date, category and status. MySQLi prepared INSERT, SELECT and UPDATE statements save and display the records. Editing without a new image keeps the current poster. Changes are also displayed on the public campus event page.

3 Poster Type and Size Validation

JPEG, PNG and WebP images are accepted up to 2 MB. The server checks actual file size, MIME type, matching extension and image contents. Image dimensions must not exceed 4096 by 4096 pixels. Text, PHP, SVG, mismatched and oversized files are rejected with clear messages.

4 Delete Confirmation

Delete opens a confirmation page before making changes. Confirmation and a valid CSRF token are required. An unregistered event is removed with a prepared DELETE statement. Events with registrations are retained and can be marked closed instead.

5 Category and Status Filters

Search matches title or venue. Category and open/closed status filters can be combined with search. The list displays the matching records or a no-results message.

6 Image Renaming and Directory Protection

Each uploaded image receives a random hexadecimal filename. Direct access to the upload directory is blocked by its Apache configuration. Posters are served through the event ID. Replaced and deleted poster files are removed after a successful database commit.

7 Audit Log and SQL Dump

Add, update and delete actions are recorded in event_audit with the event title, admin ID, time and before/after values. Database changes and audit entries use a transaction. The events-only SQL dump was exported and imported into a temporary table for verification.

Screenshots of Add Operation and Poster

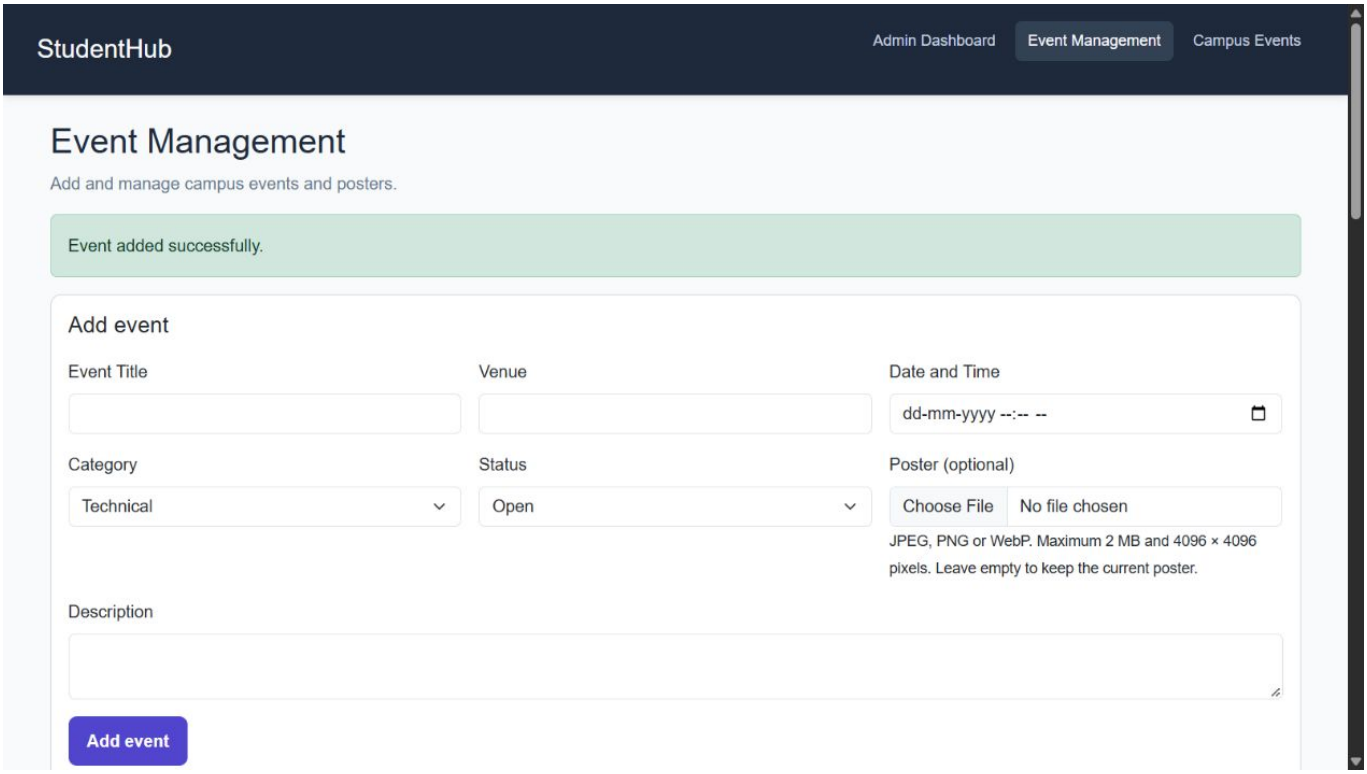


Figure 1: Event added successfully with a valid PNG poster.

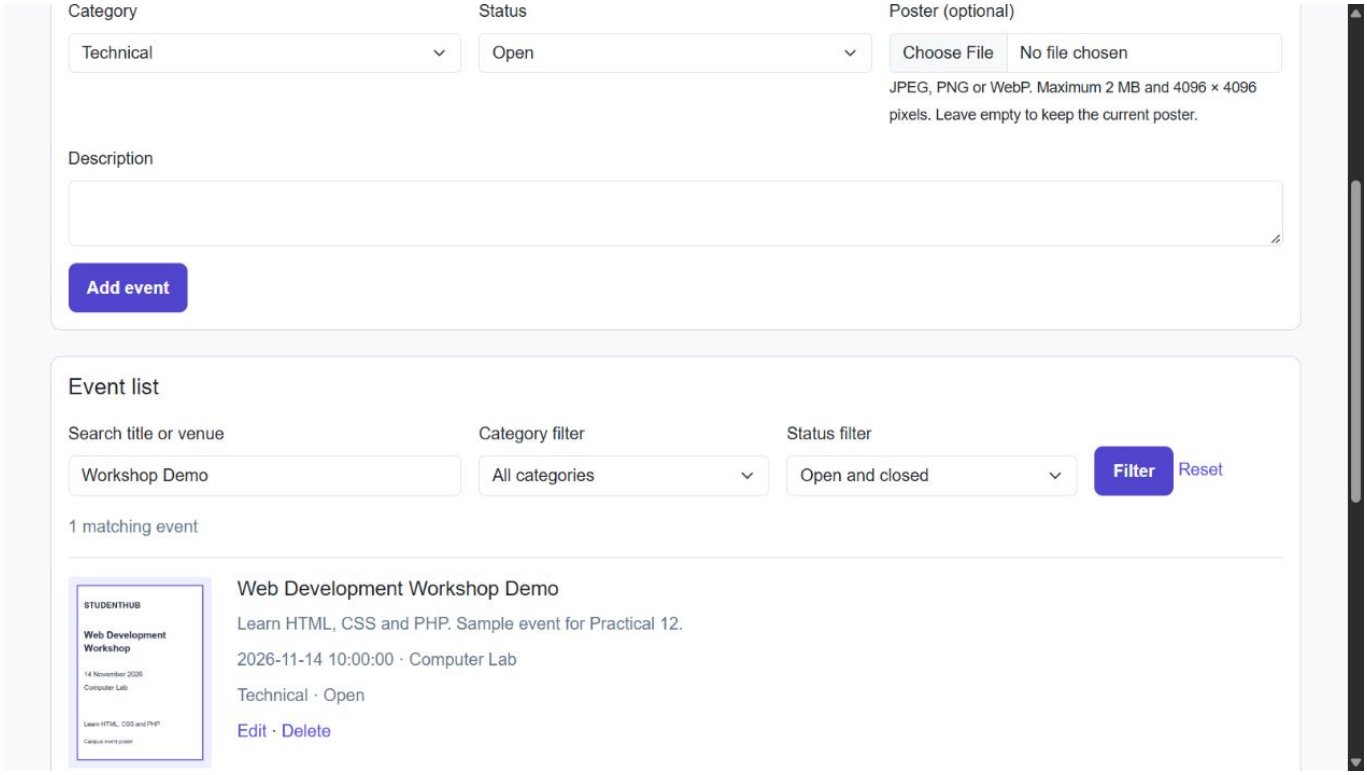


Figure 2: Saved event and poster displayed in the database event list.

Screenshots of Edit and Filters

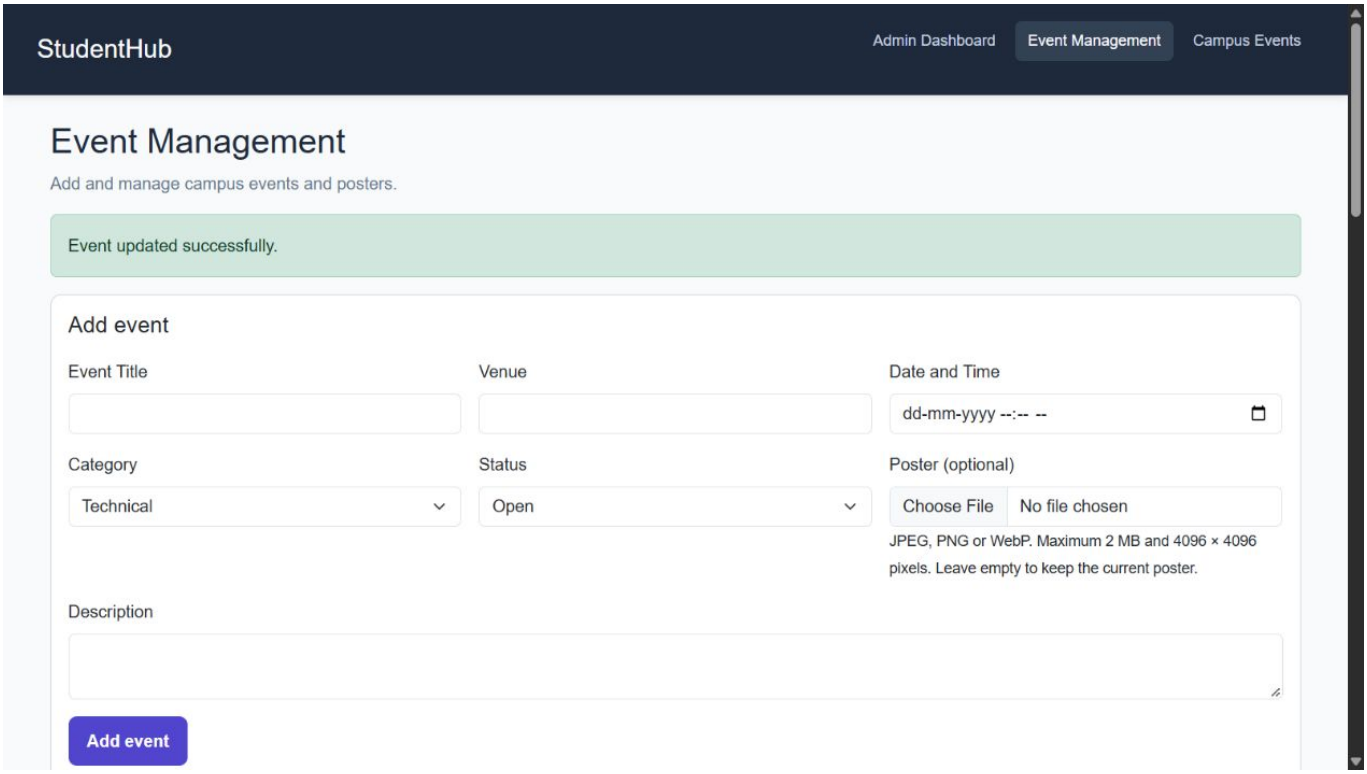


Figure 3: Event details updated successfully.

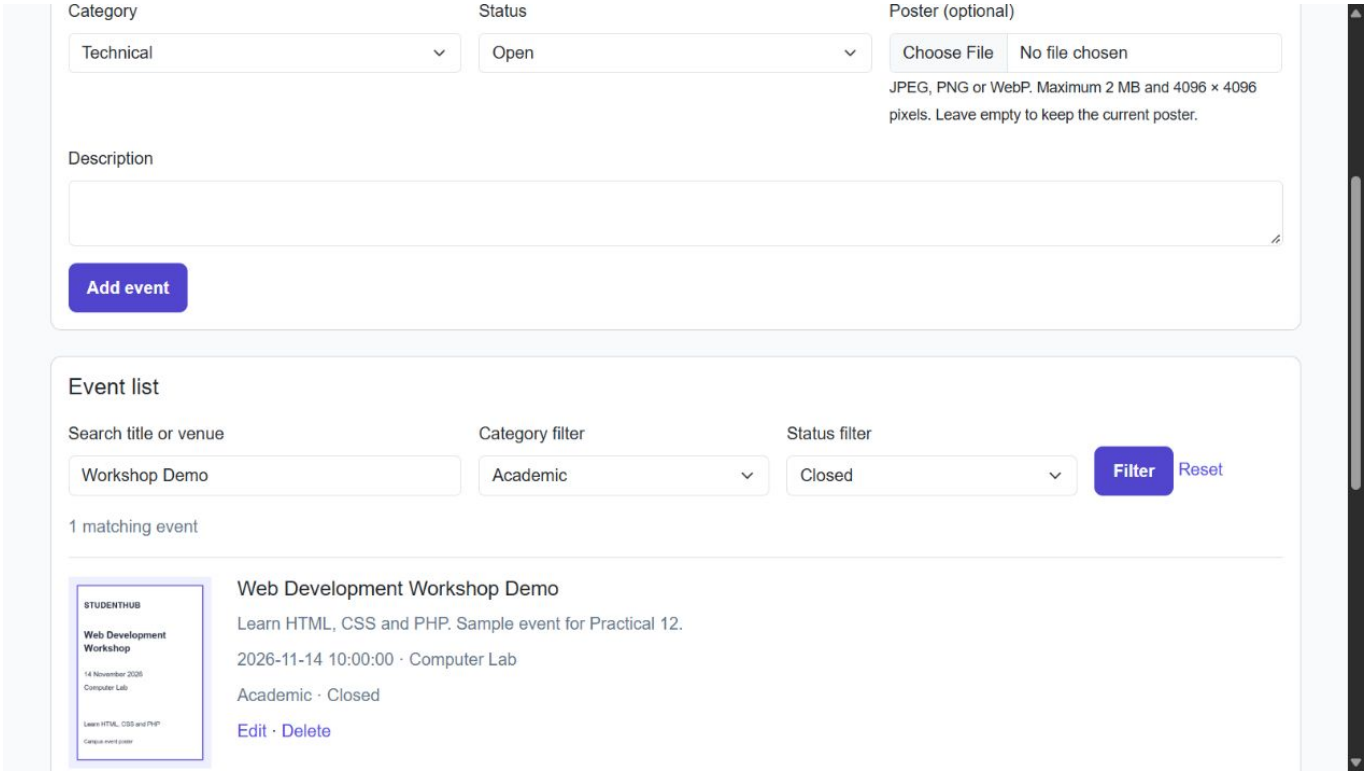


Figure 4: Search combined with academic category and closed status.

Screenshots of Upload Validation

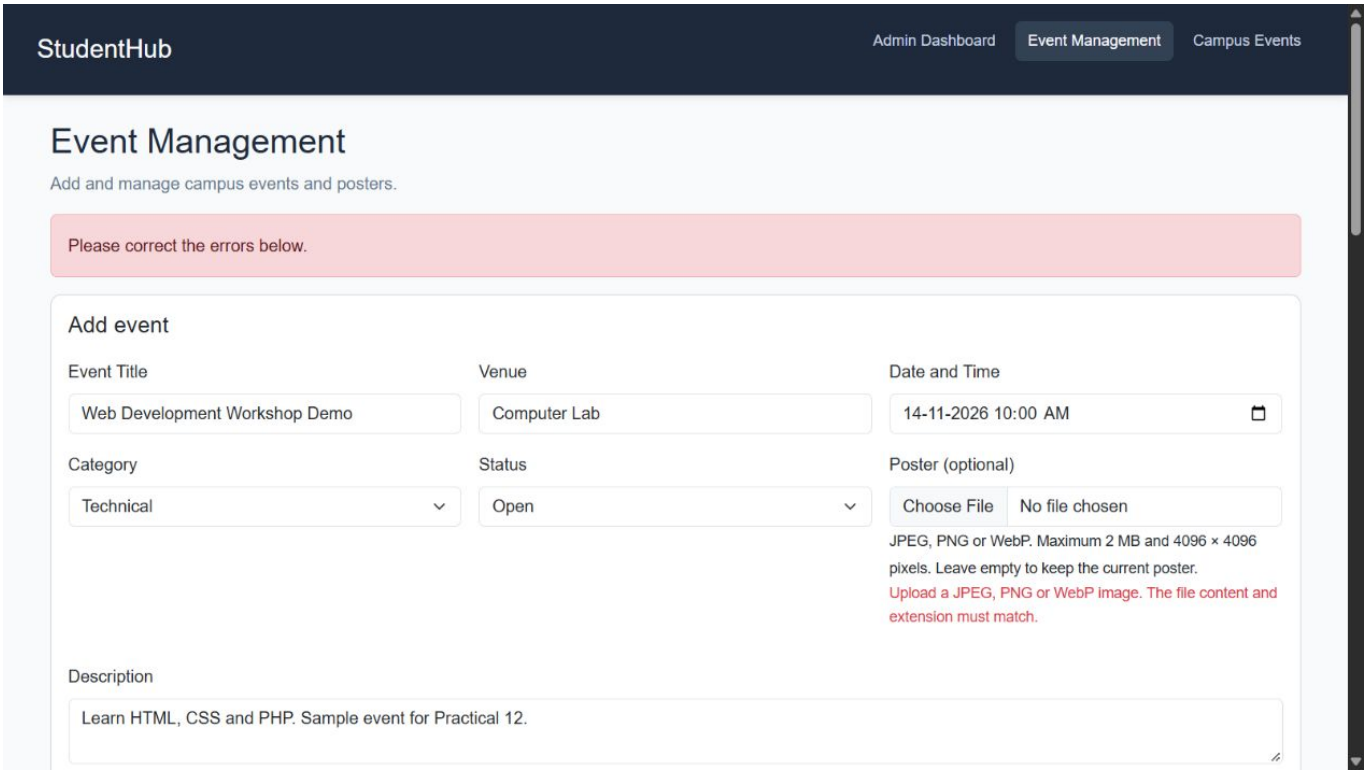


Figure 5: Unsupported poster file rejected with a validation message.

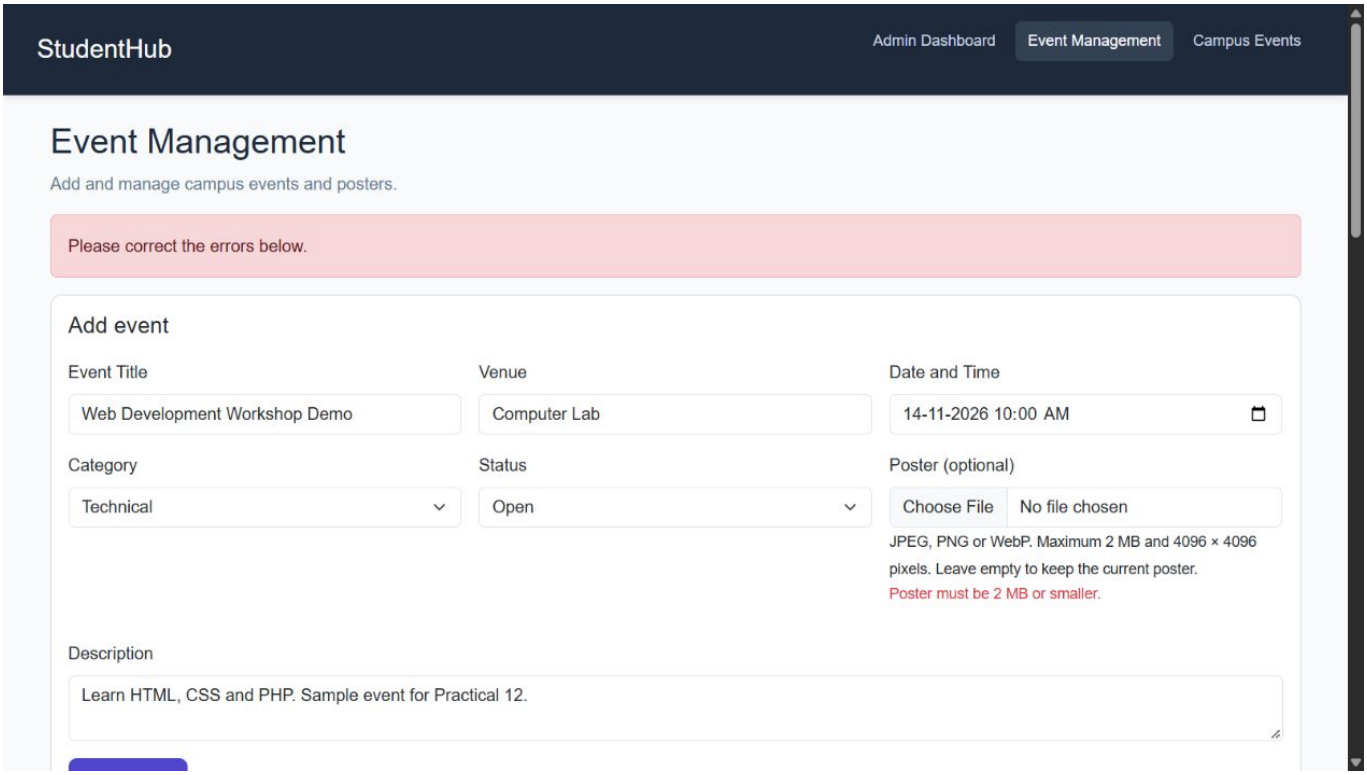


Figure 6: Poster larger than 2 MB rejected.

Screenshots of Delete Confirmation and Audit Log

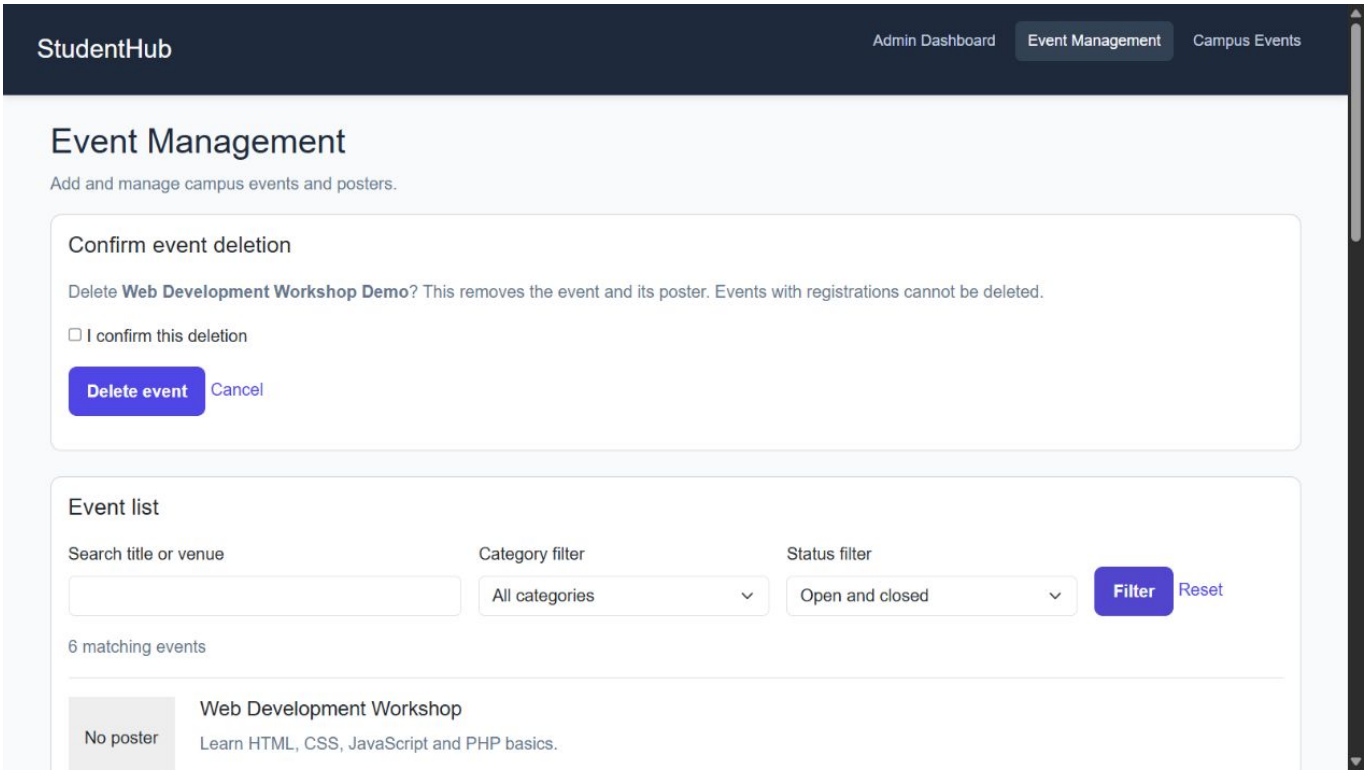


Figure 7: Confirmation is required before deleting the selected event.

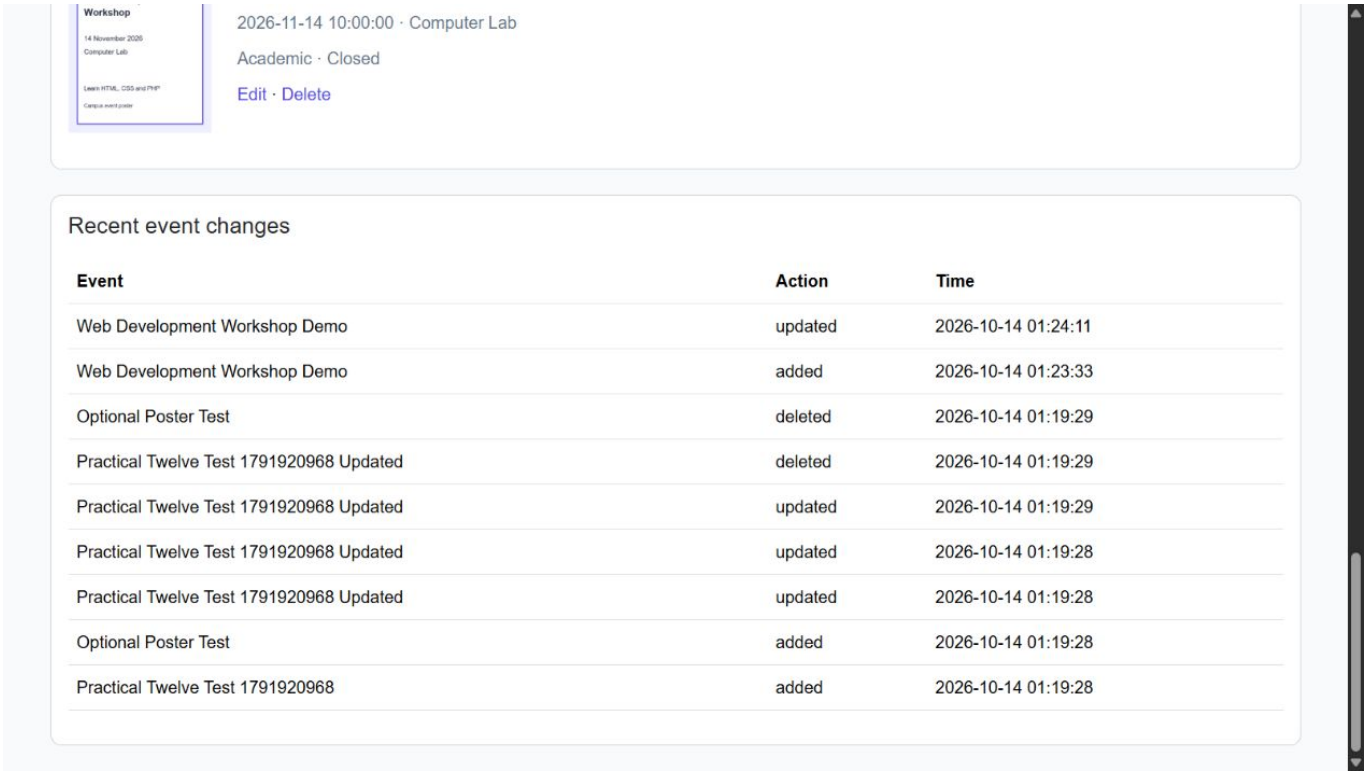


Figure 8: Audit entries show add, update and delete actions from testing.

Database Results

Check	Observed result
Records in events table and SQL dump	6
Events with posters	1
SQL dump verification	Imported successfully into a temporary table.
Audit actions	Added, updated and deleted entries recorded.

Testing Summary

Test input or action	Observed result
Admin / student / anonymous access	Allowed / HTTP 403 / login redirect.
Add with poster / no poster	Both valid events saved.
PNG, JPEG and WebP images	Accepted with random filenames.
Text, PHP, SVG and spoofed image	Rejected with validation messages.
Wrong extension / file above 2 MB	Rejected.
Image above 4096 pixels / invalid fields	Rejected.
Direct upload directory access	HTTP 403; poster endpoint works.
Edit without poster / replace poster	Old image retained / replaced image cleaned up.
Search, category and open/closed filters	Matching records displayed.
Campus event page	Updated database event returned.
Unconfirmed delete / invalid CSRF	Rejected; event retained.
Event with registrations	Deletion blocked; closing suggested.
Confirmed delete	Event and its poster removed.
Deleted event audit	Entry remains after deletion.

All 36 HTTP checks passed. Only temporary test events were deleted. Six sample events remain in the database, including the workshop demonstration event.

Key Questions and Answers

Q1 Are add, update and delete functionalities implemented correctly

Yes. Prepared INSERT, UPDATE and DELETE statements perform these operations. Valid changes were saved, deletion required confirmation and registered events were protected.

Q2 Is the UI linked with the database correctly

Yes. The admin list loads records from MySQL. Form changes update the stored records, and the campus event page also loads events from the database.

Q3 Are poster uploads validated for type and size

Yes. Actual MIME type, matching extension, file size and image contents are checked on the server. JPEG, PNG and WebP files up to 2 MB are accepted. Invalid type and size tests were rejected.

Q4 Are success and failure messages displayed

Yes. Add, update and delete actions display success messages. Invalid fields, rejected posters, missing confirmation and registered-event deletion display appropriate error messages.

GitHub Repository

https://github.com/DarshilRupani/studenthub-portal
Practical 12 source commit: 9bc9bff
Message: Complete Practical 12 event CRUD and validated poster uploads

Conclusion

The event management module was completed with CRUD operations, validated poster upload and admin access. Category and open/closed filters were added. Random image renaming, upload directory protection and audit logging were also implemented. The test cases passed and the SQL dump was verified.